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Inventor(s)	Guo; Yue et al.

Ion energy control on electrodes in a plasma reactor

Abstract

Embodiments provided herein generally include apparatus, plasma processing systems and methods for controlling ion energy distribution in a processing chamber. One embodiment of the present disclosure is directed to a method for plasma processing. The method generally includes: determining a voltage and/or power associated with a bias signal to be applied to a first electrode of a processing chamber, the voltage being determined based on a pressure inside a processing region of the processing chamber such that the voltage is insufficient to generate a plasma inside the chamber by application of the voltage and/or power to the first electrode; applying the first bias signal in accordance with the determined voltage and/or power to the first electrode; and applying a second bias signal to a second electrode of the processing chamber, wherein the second bias signal is configured to generate a plasma in the processing region and the first bias is applied while the second bias is applied.

Inventors: **Guo; Yue (Redwood City, CA), Yang; Yang (San Diego, CA), Wang; Haitao (Fremont, CA), Ramaswamy; Kartik (San Jose, CA)**

Applicant: **Applied Materials, Inc.** (Santa Clara, CA)

Family ID: **1000008780452**

Assignee: **Applied Materials, Inc. (Santa Clara, CA)**

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Primary Examiner: Allen; Joshua L

Assistant Examiner: Remavege; Christopher

Attorney, Agent or Firm: Patterson + Sheridan, LLP

Background/Summary

BACKGROUND
Field
(1) Embodiments of the present disclosure generally relate to a system used in semiconductor device manufacturing. More specifically, embodiments of the present disclosure relate to a plasma processing system used to process a substrate.
Description of the Related Art
(2) Reliably producing high aspect ratio features is one of the key technology challenges for the next generation of semiconductor devices. One method of forming high aspect ratio features uses a plasma-assisted etching process to bombard a material formed on a surface of a substrate through openings formed in a patterned mask layer formed on the substrate surface.
(3) With technology nodes advancing towards 2 nm, the fabrication of smaller features with larger aspect ratios requires atomic precision for plasma processing. For etching processes where the plasma ions play a major role, ion energy control is always challenging the development of reliable and repeatable device formation processes in the semiconductor equipment industry. In a typical plasma-assisted etching process, the substrate is positioned on an electrostatic chuck (ESC) disposed in a processing chamber, a plasma is formed over the substrate, and ions are accelerated from the plasma towards the substrate across a plasma sheath, i.e., region depleted of electrons, formed between the plasma and the surface of the substrate. Traditionally RF substrate biasing methods, which use sinusoidal RF waveforms to excite the plasma and form the plasma sheath, have bimodal ion energy distributions at the electrodes.
(4) Therefore, there is a need in the art for a source and biasing methods that are able to produce a monoenergetic ion energy peak and accurately control ion energy distributions. Accordingly, such method and technique enable the completion of a desirable plasma-assisted process on a substrate, which are critical in thin film etching and deposition applications.
SUMMARY
(5) Embodiments provided herein generally include apparatus, plasma processing systems and

methods for controlling ion energy distribution in a processing chamber.
(6) One embodiment of the present disclosure is directed to a method for plasma processing. The method generally includes: determining a voltage and/or power associated with a bias signal to be applied to a first electrode of a processing chamber, the voltage and/or power being determined based on a pressure inside a processing region of the processing chamber such that the voltage is insufficient to generate a plasma inside the chamber by application of the voltage to the first electrode; applying the first bias signal in accordance with the determined voltage to the first electrode; and applying a second bias signal to a second electrode of the processing chamber, wherein the second bias signal is configured to generate a plasma in the processing region and the first bias is applied while the second bias is applied. In some embodiments, the voltage and/or power is determined based on the pressure for a particular gap inside the processing region of the processing chamber, the gap being between either the first electrode and the second electrode or the first electrode and a grounded surface that defines at least a portion of the processing region.
(7) One embodiment of the present disclosure is directed to an apparatus for plasma processing. The apparatus generally includes: a first source voltage generator configured to apply a first bias signal to a first electrode of a processing chamber in accordance with a voltage and/or power determined based on a pressure inside a processing region of the processing chamber such that the voltage and/or power is insufficient to generate a plasma inside the chamber by application of the voltage to the first electrode; and a second source voltage generator configured to apply a second bias signal to a second electrode of the processing chamber.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) So that the manner in which the above-recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only exemplary embodiments and are therefore not to be considered limiting of its scope and may admit to other equally effective embodiments.
(2) FIG. **1** is a schematic cross-sectional view of a processing system, according to one or more embodiments, configured to practice the methods set forth herein.
(3) FIG. **2** shows a voltage waveform that is established on a substrate due to a voltage waveform applied to an electrode of a processing chamber.
(4) FIG. **3**A illustrates RF sources providing bias signals to an upper electrode and edge electrodes of a processing chamber, in accordance with certain embodiments of the present disclosure.
(5) FIG. **3**B illustrates RF sources providing bias signals to an upper electrode and a bottom electrode of a processing chamber, in accordance with certain embodiments of the present disclosure.
(6) FIG. **4** is a graph showing an ion energy distribution (IED) comparison at an electrode with different operation modes, in accordance with certain embodiments of the present disclosure.
(7) FIG. **5** is a graph showing an IED comparison at an electrode with different source power duty cycles, in accordance with certain embodiments of the present disclosure.
(8) FIG. **6** illustrates a voltage waveform as measured on an electrode and a pulsing signal provided by a source voltage generator, according to certain embodiments of the disclosure.
(9) FIG. **7**A includes curves that illustrate a desirable power range associated with a continuous wave (CW) signal provided to an electrode for use with different pressures and process chemistries, in accordance with certain embodiments of the present disclosure.
(10) FIG. **7**B illustrates Paschen's curves generated for common gases.

(11) FIG. **8** is a process flow diagram illustrating a method for plasma processing, in accordance with certain embodiments of the present disclosure.

DETAILED DESCRIPTION

(12) Some embodiments of the present disclosure are generally directed to techniques for controlling ion energy distribution (IED) at an electrode in a plasma reactor. To achieve a monoenergetic IED, a pulsed signal may be provided to an upper electrode of a processing chamber for generating a plasma inside the chamber. Moreover, a continuous wave (CW) signal may be provided to a bottom or edge electrode(s) of the processing chamber. In this manner, a monoenergetic peak may be generated, facilitating precise plasma processing, so long as the power associated with the CW signal is below a power threshold. In other words, the voltage and/or power of the CW signal is selected to be below a threshold such that the voltage and/or power of the CW signal is insufficient for generating plasma due to the application of the CW signal to avoid generating multiple IED peaks.

(13) In some embodiments, various signal parameters may be adjusted to control the IED. For example, the duty cycle of the pulsed signal may be determined based on a tradeoff between ion energy and ion energy distribution magnitude. The tradeoff is characterized by a greater IED magnitude and a lower ion energy, associated with increasing the duty cycle. Thus, the duty cycle associated with the pulsed signal provided to the upper electrode may be adjusted in accordance with this tradeoff. If a lower duty cycle is selected, resulting in a lower IED magnitude, such lower IED magnitude may be compensated for using any suitable techniques, such as increasing the power associated with the pulsed signal, increasing processing time, or increasing the pressure inside the chamber.

Plasma Processing System Examples

(14) FIG. **1** is a schematic cross-sectional view of a processing system **10** configured to perform one or more of the plasma processing methods set forth herein. In some embodiments, the processing systems **10** is configured for plasma-assisted etching processes, such as a reactive ion etch (RIE) plasma processing. However, it should be noted that the embodiments described herein may be also be used with processing systems configured for use in other plasma-assisted processes, such as plasma-enhanced deposition processes, for example, plasma-enhanced chemical vapor deposition (PECVD) processes, plasma-enhanced physical vapor deposition (PEPVD) processes, plasma-enhanced atomic layer deposition (PEALD) processes, plasma treatment processing or plasma-based ion implant processing, for example, plasma doping (PLAD) processing.

(15) As shown, the processing system **10** is configured to form a capacitively coupled plasma (CCP), where the processing chamber **100** includes an upper electrode (e.g., chamber lid **123**) disposed in a processing volume **129** facing a lower electrode (e.g., the substrate support assembly **136**) also disposed in the processing volume **129**. In a typical capacitively coupled plasma (CCP) processing system, a radio frequency (RF) source (e.g., RF generator **118**) is electrically coupled to one of the upper or lower electrode, and delivers an RF signal configured to ignite and maintain a plasma (e.g., the plasma **101**). In this configuration, the plasma is capacitively coupled to each of the upper and lower electrodes and is disposed in a processing region therebetween. Typically, the opposing one of the upper or lower electrodes is coupled to ground or to a second RF power source. In one embodiment, one or more components of the substrate support assembly **136**, such as the support base **107** is electrically coupled to a plasma generator assembly **163**, which includes the RF generator **118**, and the chamber lid **123** is electrically coupled to ground. As shown, the processing system **10** includes a processing chamber **100**, a support assembly **136**, and a system controller **126**. In some embodiments, a plasma may alternately be generated by an inductively coupled source on top of the substrate (or a peripheral coil). In this configuration, a coil may be placed on top of a ceramic lid (vacuum boundary) and powered by an RF generator **118** to generate a plasma **101** in the processing volume **129** of the processing system **10**.

(16) The processing chamber **100** typically includes a chamber body **113** that includes the chamber

lid **123**, one or more sidewalls **122**, and a chamber base **124**, which collectively define the processing volume **129**. The one or more sidewalls **122** and chamber base **124** generally include materials that are sized and shaped to form the structural support for the elements of the processing chamber **100** and are configured to withstand the pressures and added energy applied to them while a plasma **101** is generated within a vacuum environment maintained in the processing volume **129** of the processing chamber **100** during processing. In one example, the one or more sidewalls **122** and chamber base **124** are formed from a metal, such as aluminum, an aluminum alloy, or a stainless steel alloy.

(17) A gas inlet **128** disposed through the chamber lid **123** is used to deliver one or more processing gases to the processing volume **129** from a processing gas source **119** that is in fluid communication therewith. A substrate **103** is loaded into, and removed from, the processing volume **129** through an opening (not shown) in one of the one or more sidewalls **122**, which is sealed with a slit valve (not shown) during plasma processing of the substrate **103**.

(18) The system controller **126**, also referred to herein as a processing chamber controller, includes a central processing unit (CPU) **133**, a memory **134**, and support circuits **135**. The system controller **126** is used to control the process sequence used to process the substrate **103**, including the substrate biasing methods described herein. The CPU **133** is a general-purpose computer processor configured for use in an industrial setting for controlling the processing chamber and sub-processors related thereto. The memory **134** described herein, which is generally non-volatile memory, may include random access memory, read-only memory, floppy or hard disk drive, or other suitable forms of digital storage, local or remote. The support circuits **135** are conventionally coupled to the CPU **133** and comprise cache, clock circuits, input/output subsystems, power supplies, and the like, and combinations thereof. Software instructions (program) and data can be coded and stored within the memory **134** for instructing a processor within the CPU **133**. A software program (or computer instructions) readable by CPU **133** in the system controller **126** determines which tasks are performable by the components in the processing system **10**.

(19) Typically, the program, which is readable by CPU **133** in the system controller **126**, includes code, which, when executed by the processor (CPU **133**), performs tasks relating to the plasma processing schemes described herein. The program may include instructions that are used to control the various hardware and electrical components within the processing system **10** to perform the various process tasks and various process sequences used to implement the methods described herein. In one embodiment, the program includes instructions that are used to perform one or more of the operations described below in relation to FIG. **6**.

(20) The processing system may include a plasma generator assembly **163**, a first source assembly **196** for establishing a first PV waveform at a bias electrode **104**. In some embodiments, the plasma generator assembly **163** delivers an RF signal to the support base **107** (e.g., power electrode or cathode) which may be used to generate (maintain and/or ignite) a plasma **101** in a processing region disposed between the substrate support assembly **136** and the chamber lid **123**. In some embodiments, the RF generator **118** is configured to deliver an RF signal having a frequency that is greater than 1 MHz or more, or about 2 MHz or more, such as about 13.56 MHz or more. In one example, the RF generator **118** is configured to deliver an RF signal having a frequency that is between about 13.56 MHz and about 200 MHz, such as between about 40 MHz and about 60 MHz.

(21) As discussed above, in some embodiments, the plasma generator assembly **163**, which includes an RF generator **118** and an RF generator assembly **160**, is generally configured to deliver a desired amount of a continuous wave (CW) or pulsed RF power at a desired substantially fixed sinusoidal waveform frequency to a support base **107** of the substrate support assembly **136** based on control signals provided from the system controller **126**. During processing, the plasma generator assembly **163** is configured to deliver RF power (e.g., an RF signal) to the support base **107** disposed proximate to the substrate support **105**, and within the substrate support assembly **136**. The RF power delivered to the support base **107** is configured to ignite and maintain a

processing plasma **101** containing processing gases disposed within the processing volume **129**. Configurations that provide RF power to the support base **107** to generate the plasma **101** will typically include a configuration where the chamber lid **123** is grounded.

(22) In some embodiments, the support base **107** is an RF electrode that is electrically coupled to the RF generator **118** via an RF matching circuit **162** and a first filter assembly **161**, which are both disposed within the RF generator assembly **160**. The first filter assembly **161** includes one or more electrical elements that are configured to substantially prevent a current generated by the output of a waveform generator **150** from flowing through an RF power delivery line **167** and damaging the RF generator **118**. The first filter assembly **161** acts as a high impedance (e.g., high Z) to the waveform generated by the waveform generator **150**, and thus inhibits the flow of current to the RF matching circuit **162** and RF generator **118**.

(23) In some embodiments, the RF generator assembly **160** and RF generator **118** are used to ignite and maintain a processing plasma **101** using the processing gases disposed in the processing volume **129** and fields generated by the RF power (RF signal) delivered to the support base **107** by the RF generator **118**. The processing volume **129** is fluidly coupled to one or more dedicated vacuum pumps through a vacuum outlet **120**, which maintain the processing volume **129** at sub-atmospheric pressure conditions and evacuate processing and/or other gases, therefrom. In some embodiments, the substrate support assembly **136**, disposed in the processing volume **129**, is disposed on a support shaft **138** that is grounded and extends through the chamber base **124**. However, in some embodiments, the RF generator assembly **160** is configured to deliver an RF power to the bias electrode **104** disposed in the substrate support **105** versus the support base **107**.

(24) The substrate support assembly **136**, as briefly discussed above, generally includes the substrate support **105** (e.g., ESC substrate support) and support base **107**. In some embodiments, the substrate support assembly **136** can additionally include an insulator plate **111** and a ground plate **112**, as is discussed further below. The support base **107** is electrically isolated from the chamber base **124** by the insulator plate **111**, and the ground plate **112** is interposed between the insulator plate **111** and the chamber base **124**. The substrate support **105** is thermally coupled to and disposed on the support base **107**. In some embodiments, the support base **107** is configured to regulate the temperature of the substrate support **105**, and the substrate **103** disposed on the substrate support **105**, during substrate processing.

(25) Typically, the substrate support **105** is formed of a dielectric material, such as a bulk sintered ceramic material, such as a corrosion-resistant metal oxide or metal nitride material, for example, aluminum oxide (Al.sub.2O.sub.3), aluminum nitride (AlN), titanium oxide (TiO), titanium nitride (TiN), yttrium oxide (Y.sub.2O.sub.3), mixtures thereof, or combinations thereof. In embodiments herein, the substrate support **105** further includes the bias electrode **104** embedded in the dielectric material thereof. In some embodiments, one or more characteristics of the RF power used to maintain the plasma **101** in the processing region over the bias electrode **104** are determined and/or monitored by measuring an RF waveform established at the bias electrode **104**.

(26) In one configuration, the bias electrode **104** is a chucking pole used to secure (i.e., chuck) the substrate **103** to the substrate supporting surface **105**A of the substrate support **105** and to bias the substrate **103** with respect to the processing plasma **101** using one or more of the pulsed-voltage biasing schemes described herein. Typically, the bias electrode **104** is formed of one or more electrically conductive parts, such as one or more metal meshes, foils, plates, or combinations thereof.

(27) In some embodiments, the bias electrode **104** is electrically coupled to a clamping network **116**, which provides a chucking voltage thereto, such as static DC voltage between about −5000 V and about 5000 V, using an electrical conductor, such as the coaxial power delivery line **106** (e.g., a coaxial cable). As will be discussed further below, the clamping network **116** includes ESC clamping voltage compensation circuit elements **116**A, a DC power supply **155**, and a ESC clamping voltage compensation module blocking capacitor, which is also referred to herein as the

blocking capacitor C.sub.5. The blocking capacitor C.sub.5 is disposed between the output of a pulsed voltage (PV) waveform generator **150** and the bias electrode **104**.
(28) The substrate support assembly **136** may further include the edge control electrode **115** that is positioned below the edge ring **114** and surrounds the bias electrode **104** and/or is disposed a distance from a center of the bias electrode **104**. In general, for a processing chamber **100** that is configured to process circular substrates, the edge control electrode **115** is annular in shape, is made from a conductive material, and is configured to surround at least a portion of the bias electrode **104**. In some embodiments, such as shown in FIG. **1**, the edge control electrode **115** is positioned within a region of the substrate support **105**. In some embodiments, as illustrated in FIG. **1**, the edge control electrode **115** includes a conductive mesh, foil, and/or plate that is disposed a similar distance (i.e., Z-direction) from the substrate supporting surface **105**A of the substrate support **105** as the bias electrode **104**. In some other embodiments, the edge control electrode **115** includes a conductive mesh, foil, and/or plate that is positioned on or within a region of a quartz pipe **110**, which surrounds at least a portion of the bias electrode **104** and/or the substrate support **105**. Alternately, in some other embodiments (not shown), the edge control electrode **115** is positioned within or is coupled to the edge ring **114**, which is disposed on and adjacent to the substrate support **105**. In this configuration, the edge ring **114** is formed from a semiconductor or dielectric material (e.g., AlN, etc.).
(29) A power delivery line **157** electrically connects the output of the waveform generator **150** of the first source assembly **196** to an optional filter assembly **151** and the bias electrode **104**. While the discussion below primarily discusses the power delivery line **157** of the first source assembly **196**, which is used to couple a waveform generator **150** to the bias electrode **104**, the power delivery line **158** of the second source assembly **198**, which couples a waveform generator **150** to an upper electrode (e.g., chamber lid **123**), will include the same or similar components. The electrical conductor(s) within the various parts of the power delivery line **157** may include: (a) one or a combination of coaxial cables, such as a flexible coaxial cable that is connected in series with a rigid coaxial cable, (b) an insulated high-voltage corona-resistant hookup wire, (c) a bare wire, (d) a metal rod, (e) an electrical connector, or (f) any combination of electrical elements in (a)-(e). The optional filter assembly **151** includes one or more electrical elements that are configured to substantially prevent a current generated by the output of the RF generator **118** from flowing through the power delivery line **157** and damaging the waveform generator **150**. The optional filter assembly **151** acts as a high impedance (e.g., high Z) to RF signal generated by the RF generator **118**, and thus inhibits the flow of current to the waveform generator **150**.
(30) In some embodiments, a CW signal generator **191** may be included within a waveform generator **150** to generate a CW signal at the output thereof. The CW signal, such as a sinusoidal RF signal, may be applied to one or more electrodes of the processing chamber **100**, such as the electrode **104**, edge electrode **115**, or any combination thereof. In one example, the CW signal generator **191** is configured to deliver an RF signal having a frequency that is greater than 1 MHz, such as between about 13.56 MHz and about 200 MHz, such as between about 40 MHz and 60 MHz.
(31) In some embodiments, the processing chamber **100** further includes the quartz pipe **110**, or collar, that at least partially circumscribes portions of the substrate support assembly **136** to prevent the substrate support **105** and/or the support base **107** from contact with corrosive processing gases or plasma, cleaning gases or plasma, or byproducts thereof. Typically, the quartz pipe **110**, the insulator plate **111**, and the ground plate **112** are circumscribed by a liner **108**. In some embodiments, a plasma screen **109** is positioned between the cathode liner **108** and the sidewalls **122** to prevent plasma from forming in a volume underneath the plasma screen **109** between the liner **108** and the one or more sidewalls **122**.
(32) FIG. **2** illustrates an example voltage waveform **200** established at a substrate in a processing chamber (e.g., processing chamber **100**). In this example, the waveform **200** is generated due to the

application of a waveform by the waveform generator **150** of the second source assembly **198** to one of the electrodes within the processing chamber **100**, such as the bias electrode **104** and/or edge electrode **115**. The waveform **200** includes an ion current stage and a sheath collapse stage, as shown. At the beginning of the ion current stage, a drop of substrate voltage, which is created by the falling edge **204**, creates a high voltage sheath to form above the substrate, accelerating positive ions to the substrate. The positive ions that bombard the surface of the substrate during the ion current stage deposit a positive charge on the substrate surface, which, if uncompensated, causes a gradual increase of the substrate voltage positively (i.e., positive slope during phase **205** of the voltage waveform **200**) during the ion current stage, as shown. However, the uncontrolled accumulation of positive charge on the substrate surface undesirably gradually discharges the sheath and chuck capacitors, slowly decreasing the sheath voltage drop and bringing the substrate potential closer to zero. The accumulation of positive charge results in the voltage droop (i.e., positive slope during phase **205**) in the voltage waveform established at the substrate. The voltage difference between the beginning and end of the ion current phase determines an ion energy distribution function (IEDF) width. The greater the voltage difference, the wider the IEDF width.
(33) During a portion of the voltage waveform, plasma bulk electrons are attracted to the substrate surface due to the rising edge **202** of the pulse step, but those electrons cannot establish a negative DC sheath potential yet as there are equal amounts of positive charge on the electrode (e.g., electrode **104**). The substrate and the dielectric disposed between the electrode and the substrate supporting surface **105**A form a capacitor, which has an effective capacitance C.sub.esc, which will allow an equal amount of positive charge on the electrode to cancel the field generated by the electrons disposed on the substrate surface. At the falling edge **204** of the pulse step, the positive charge on the electrode is neutralized by the electrons from the waveform generator, and therefore, a negative DC voltage is established on the substrate surface. If the formed DC voltage is held constant, then mono-energy ion bombardment is achieved. The negative DC voltage (Vdc) can be approximated by using the magnitude of the falling edge (ΔV) and the ratio between the C.sub.esc and the sheath capacitance C.sub.sheath in accordance with the following equation:
Vdc=*ΔV***C*.sub.esc/(*C*.sub.esc*C*.sub.sheath)
Technique for Waveform Generation
(34) As device dimensions are scaling down below 5 nm, accurate ion energy control is becoming more important in thin film etching and deposition applications. The aspects described herein present methods and systems for ion energy control on one or more electrodes in a plasma reactor. In some embodiments, an ion energy control method is adopted with an active control module, in order to achieve selectivity, film quality and edge profile control, especially at or near an edge of the substrate disposed within the plasma chamber. Some embodiments facilitate achieving a narrow width monoenergetic ion energy distribution (IED) on electrodes. The position and width of the peak energy can also be controlled precisely, as described in more detail herein.
(35) FIG. **3**A illustrates RF sources providing bias signals to an upper electrode **304** (e.g., corresponding to chamber lid **123** of FIG. **1**), bias electrode(s) **104** and edge electrodes (e.g., edge electrode **115** of FIG. **1**) of a processing chamber, in accordance with certain embodiments of the present disclosure. For example, a source RF generator **302** (e.g., corresponding to the source assembly **198**) may generate a pulsed signal provided to the upper electrode **304**. In some embodiments, a RF source power is applied via the source RF generator **302** to the upper electrode (e.g., chamber lid **123**) for plasma production. The frequency of the RF source power may be from 13.56 MHz to a high frequency band such as around 200 MHz. As a few examples, the frequency of the RF source power may be 60 MHz, 120 MHz, or 162 MHz. The pulsing frequency of the source power may be from 100 Hz to 5 kHz and the duty cycle associated with the pulsing may range from 5% to 95%.
(36) In some examples, the RF source power can also be delivered via a bottom electrode (e.g., electrode **104**) by use of the waveform generator **150**. For example, the voltage generator **306** may

be part of the first source assembly **196** described with respect to FIG. **1**, and may provide a RF source power to the bottom electrode **310** (e.g., corresponding to electrode **104** of FIG. **1**). For example, the voltage generator **306** may apply a bias power the bottom electrode **310** (e.g., electrode **104** of FIG. **1**) with a frequency range from 100 kHz to 60 MHz. The bias power may be operated in either a continuous or a pulsed mode. In one example, an RF waveform is provided to the electrode **310** by the first source assembly **196**, and thus is different from the voltage waveform **200** discussed above in relation to FIG. **2**. The RF waveform may be provided by the CW signal generator **191**, which is discussed above.

(37) In some embodiments of the present disclosure, a CW signal generator **308** may generate a continuous wave (CW) bias signal which may be provided to the edge electrode **115**, as shown. The CW signal generator **308** may include the CW signal generator **191** described with respect to FIG. **1**. The CW signal allows control of ion energy distribution on edge electrodes in a plasma processing chamber. In other words, a low-frequency RF power in the frequency range of 50 kHz to 2 MHz may be delivered to one or more edge electrodes and sourced in a continuous mode. That is, the bias signals applied to the one or more edge electrodes may be RF type continuous wave (CW) signals.

(38) While the discussion herein primarily discloses the use of RF sources that are configured to provide a continuous wave (CW) signal (e.g., CW bias signal), such as the generators **306** or **308**, this configuration is not intended to be limiting as to the scope of the disclosure since the RF sources could also be replaced by a source that supplies a pulse voltage waveform. In one example, one or more of the generators **306** and **308** can be configured to deliver, or replaced by a waveform generator that is configured to deliver, a pulsed voltage waveform similar to the waveform **200** illustrated in FIG. **2**.

(39) In order to achieve a monoenergetic ion energy distribution, the signal applied to the edge electrode may be set to be within a power range, as described in more detail herein. For example, the power of a bias signal applied to an electrode may be set within a voltage and/or power range (e.g., below a power threshold) such that the bias voltage and/or power is insufficient for plasma excitation due to the application of the CW bias signal, or pulsed voltage waveform. Surprisingly, it has been found that a monoenergetic ion energy distribution can be achieved at an electrode, such as the edge electrode in this process regime. Setting the power to be below the power threshold may involve setting the voltage of the CW bias signal to be below a voltage threshold, in some embodiments. In some embodiments in which a pulsed waveform is used, the maximum voltage provided during the delivery of the pulsed waveform to the electrode during processing, or peak voltage V.sub.p (FIG. **2**) of the pulsed waveform, is maintained below the desired threshold voltage.

(40) In the configuration illustrated in FIG. **3**A, a CW bias signal is provided to the edge electrode **115** via the CW signal generator **308** and the voltage generator **306** is configured and adapted to apply a bias signal to the bottom electrode **310**. In one embodiment, the voltage generator **306** is configured to apply a pulsed voltage waveform to the bias electrode **104** and the CW signal generator **308** is configured to apply a CW bias signal to the edge electrode **115**. In another embodiment, the voltage generator **306** is configured to apply an RF waveform to the bias electrode **104** and the CW signal generator **308** is configured to apply a bias signal to the edge electrode **115**, such as a CW bias signal. In another embodiment, the voltage generator **306** is configured to apply a bias signal that includes a pulsed voltage waveform to the bias electrode **104** and the CW signal generator **308** is configured to apply a bias signal that includes a pulsed voltage waveform to the edge electrode **115**. In yet another embodiment, the voltage generator **306** is configured to apply a bias signal that includes a CW bias signal to the bias electrode **104** and the CW signal generator **308** is configured to apply a bias signal that includes a pulsed voltage waveform signal to the edge electrode **115**. In any of these possible embodiments, the power of the CW bias signal or the pulsed voltage waveform may be set within a voltage and/or power range (i.e., below a voltage or power

threshold) such that the bias power is insufficient for plasma excitation due to the application of the CW bias signal or the pulsed voltage waveform.

(41) FIG. **3**B illustrates voltage sources providing bias signals to an upper electrode and a bottom electrode of a processing chamber, allowing for monoenergetic ion energy distribution control in a plasma processing chamber. As shown, source RF generator **302** may provide the RF source power to the upper electrode **304** for plasma production. In some examples, the RF source power may be provided to a bottom electrode (e.g., bias electrode **104** or support base **107**). In other examples, the RF source power may be provided to an inductively coupled plasma (ICP) source electrode.

(42) In the configuration illustrated in FIG. **3**B, a CW bias signal may be provided to the bottom electrode **310** (e.g., bias electrode **104** or support base **107** of FIG. **1**) via the CW signal generator **308**. The power of the CW bias signal, or pulsed voltage waveform, may be set within a power range such that the bias power is insufficient for plasma excitation due to the application of the bias signal. A monoenergetic ion energy distribution can be achieved at the bottom electrode in this process regime. As used herein, an electrode that is provided a CW bias signal, or pulsed voltage waveform, is generally referred to as a CW electrode, such as the edge electrode **115** or the bottom electrode **310**. For ease of discussion, while a bias signal applied to an electrode is primarily referred to below as a CW bias signal, it should be understood that the CW bias signal could be replaced by a pulsed voltage waveform.

(43) FIG. **4** is a graph **400** showing IEDs at a CW electrode with different operation modes. When all RF powers (source and bias) are operated in a CW mode, IED **402** measured on the CW electrode has a low energy peak. A preferable single peak IED **404** at a higher energy is achieved on the CW electrode when the source power (e.g., from the source RF generator **302**) is pulsed. For comparison purposes, RF signal from the source RF generator that was used to generate the lower energy IED peak, was maintained at a first power level and in continuous mode (100% duty cycle), whereas RF signal, used to generate the higher IED peak, was provided at the same power level and operated in the pulsed mode with a duty cycle from 0% to 100%. One will note that the IED shifts towards higher energy if a higher bias power is used at the CW electrode.

(44) FIG. **5** is a graph **500** showing an IED comparison at the CW electrode with different pulsed source power duty cycles. As described, the source power from the source RF generator **302** may be pulsed. The pulsed signal may be set to different duty cycles (e.g., 25%, 50%, and 75%). A narrower IED on the CW electrode may be achieved with a longer source power duty cycle. In other words, the IED (e.g., associated with peak **502**) is narrower for the 75% duty cycle as compared to the IED (e.g., associated with peak **506**) for the 25% duty cycle. On the other hand, a higher ion energy single peak **506** can be desirably achieved with a shorter source power duty cycle (e.g., 25% duty cycle).

(45) FIG. **6** illustrates a voltage waveform **602** as measured on the CW electrode and a pulsing signal **604** (e.g., pulsed RF signal) provided by the source RF generator **302** to an electrode, such as the electrode **304**. As shown, the source power is in pulsed mode and CW bias power is in CW mode. When the source power is off, the voltage on the CW electrode becomes increasingly negative, resulting in a high energy monoenergetic IED. In other words, the longer the pulsing signal **604** is off, the more negative the voltage on the CW electrode will become, resulting in a higher energy as shown by graph **500** of FIG. **5**. Therefore, as shown in FIG. **5**, for a 25% duty cycle pulsing signal (e.g., where the pulsing signal is off for 75% of the period of the pulsing signal), the resulting peak is associated with a higher energy yet has a lower IED magnitude.

(46) The power supplied to the CW electrode may be set to a specific constant power to implement a monoenergetic IED. As shown in FIG. **5**, the peak **502** associated with a 75% duty cycle has a higher magnitude than peak **504** associated with a 50% duty cycle, and the peak **504** has a higher magnitude than peak **506** associated with a 25% duty cycle. To implement a monoenergetic IED, the power supplied by the CW bias signal generator may be set to be below a certain power threshold (or voltage threshold). The threshold may be set based on various factors such as the

pressure, gap (e.g., gap **350** in FIGS. **3**A-**3**B) between electrodes, and chemistries associated with the chamber, as described in more detail with respect to FIG. **7**A.

(47) FIG. **7**A is an example of a graph **700** that includes one or more power range curves **701**A-**701**B that illustrate a power range associated with the CW bias signal under different process pressures and using different process chemistries for a given processing chamber **100** configuration. The power supplied to the CW electrode is within a range where RF power is insufficient for plasma excitation. In one embodiment, as illustrated in FIG. **7**A, the graph **700** includes two curves that each illustrate the maximum power that can be applied to an electrode for a given process chemistry as a function of processing pressure that can be applied to the electrode in a known chamber configuration and not generate a plasma. In alternate embodiments, the graph **700** may include a curve that illustrates a maximum voltage that can be applied to an electrode for a known process chemistry as a function of processing pressure that can be applied to the electrode in a known chamber configuration and not generate a plasma. As shown in FIG. **7**A, a higher RF power can be applied to the CW electrode at a lower pressure to avoid the generation of a plasma by the delivery of the CW bias signal. Electronegative process chemistries, for example argon (Ar)/chlorine (Cl.sub.2)/dioxygen (O.sub.2), can operate at higher powers before plasma excitation occurs, as compared to the Ar chemistries. Moreover, the pressure range may be from 1 mTorr to 500 mTorr. The power threshold (e.g., voltage threshold) that is insufficient for plasma excitation may also depend on a gap formed between the CW electrode and a grounded surface or counter electrode disposed in the chamber. For example, referring back to FIG. **3**B, a gap **350** may exist between a CW electrode (e.g., electrode **310** or electrode **115**) to which a CW signal is applied and the upper electrode **304** to which the source power is applied. Thus, the CW signal's power (or voltage) may be determined for a particular gap **350** as described. In some embodiments, the gap **350** may be adjusted prior to or during processing by use of an actuator (not shown), or other adjustable mounting hardware, that is configured to position the CW electrode at a desired location within the processing volume.

(48) In some embodiments, the limit on the power provided to a CW electrode to avoid the generation of a plasma due to the delivery of CW signal can be determined for a given process chemistry and chamber configuration by experimentally generating a Paschen's curve over a desired process regime and then maintaining the process parameters during processing at a level below a point where a plasma will be generated. An example of five different Paschen's curves generated for common gases is illustrated in FIG. **7**B. However, the characteristics of a Paschen's curve depend on the process chemistry and chamber configuration, and thus will need to be generated as one or more of the chamber configuration and process chemistry parameters are varied. It is believed that by staying below a voltage level at a given pressure times gap distance (e.g., "pd" in Torr-cm in FIG. **7**B) that is defined by a generated Paschen's curve for a given gas chemistry one can be assured that an arc and thus plasma will not be generated due to the signal provided to the CW electrode.

(49) Based on the creation of curves that are the same as or similar to the ones illustrated in FIG. **7**A, or generated Paschen's curve similar to the ones shown in FIG. **7**B, the system controller **126** for a given plasma processing recipe (e.g., CW provided power, gas composition, chamber configuration) that is to be executed during a plasma processing process can be used to control and/or limit the power provided to the CW electrode to assure that the applied voltage and/or power stays below a level that will generate a plasma in the processing volume. As discussed above, the controlled or limited voltage and/or power that is applied can be provided by the delivery of a CW bias signal or a pulsed voltage waveform from the CW signal generator **308**. However, the controlled or limited voltage and/or power that is applied can also be provided by one or more of the generators **306** and **308**.

(50) In an effort to desirably control the results of a plasma process, the duty cycle of the source power (e.g., the pulsed signal generated by the source RF generator **302**) may be determined based

on a tradeoff between ion energy and ion energy distribution magnitude. The tradeoff is characterized by a greater ion energy distribution magnitude and a lower ion energy, associated with increasing the duty cycle, as described with respect to FIG. **5**. For example, a lower duty cycle (e.g., 25%) may be selected to obtain a higher energy, yet with a tradeoff of a lower ion energy magnitude, as shown by graph **500**.

(51) The lower duty cycle magnitude may be compensated for using any suitable technique. Such a technique may include increasing the source power (e.g., power of pulsed signal generated by the source RF generator **302**), increasing processing time, increasing the pressure inside the chamber, or any combination thereof. Embodiments described herein allow for a narrow width monoenergetic IED on the CW electrodes for uniformity control. The single IED peak can be controlled by power delivered to the CW electrode and the duty cycle of the pulsed source power.

(52) FIG. **8** is a process flow diagram illustrating a method **800** for plasma processing. Method **800** may be performed by a processing system. The processing system may include voltage generation components (e.g., source assembly **196**, **198** and CW signal generator **191**) and a system controller such as the system controller **126**.

(53) Method **800** begins, at activity **802**, with the processing system determining a voltage and/or power associated with a bias signal to be applied to a first electrode of a processing chamber. In one configuration, the first electrode is a CW electrode, such as the bias electrode **104**, edge electrode **115**, or any combination thereof. The voltage and/or power may be determined based on the chamber configuration, gas composition and a pressure inside a processing region of the processing chamber such that the voltage is insufficient to generate a plasma inside the chamber by application of the voltage to the first electrode. In some embodiments, the voltage and/or power can be determined by comparing process recipe parameters (e.g., gas composition, process pressure set point(s), and electrode spacing(s) (e.g., gap)) with one or more of the curves illustrated in FIGS. **7**A-**7**B for a desired process recipe and known chamber configuration. The comparison may be completed by comparing curve related information (e.g., power range curve (e.g., FIG. **7**A) or Paschen's curve (e.g., FIG. **7**B)) stored in the memory of the system controller **126** with the desired process recipe parameters.

(54) At activity **804**, the processing system applies the first bias signal (e.g., a CW signal) in accordance with the determined voltage and/or power to the first electrode, and at activity **806**, applies a second bias signal (e.g., source power generated via source RF generator **302**) to a second electrode of the processing chamber, where the second bias signal is configured to generate a plasma in the processing region and the first bias is applied while the second bias is applied. In some configurations, the second electrode is the upper electrode **304** (FIGS. **3**A-**3**B) or the support base **107**. The voltage or power applied to the first electrode is determined (e.g., at activity **802**) based on the pressure for a particular gap formed inside the processing region of the processing chamber. The gap can be formed between the first electrode and the second electrode in a configuration where the first electrode and second electrode oppose each other, as illustrated in FIG. **3**A-**3**B, or formed between the first electrode and a grounded surface used to define at least a portion of the processing volume **129** of the processing chamber **100** (e.g., chamber lid **123**).

(55) In some embodiments, the processing system further applies a third bias signal (e.g., a CW signal) to a third electrode of the processing chamber. The first electrode and the third electrode may be positioned within a substrate support assembly disposed within the processing region and a distance from a substrate supporting surface of the substrate support assembly. For example, the first electrode and the third electrode may be the edge electrode (e.g., edge electrode **115**) and the chucking electrode (e.g., bias electrode **104**), respectively.

(56) In some embodiments, the voltage and/or power is further determined based on a gas composition disposed within the processing region of the processing chamber. As described herein, the voltage and/or power may be determined to be below a voltage and/or power threshold such that the voltage is insufficient to generate a plasma inside the chamber. For example, the voltage

and/or power may be determined to be below a voltage and/or power threshold such that a monoenergetic peak is generated for the plasma processing.

(57) In some embodiments, the second bias signal is a pulsed signal having an on period and an off period (e.g., as shown in FIG. **6**). A duty cycle associated with the pulsed signal may be determined based on a tradeoff between ion energy and ion energy distribution magnitude associated with adjustments to the duty cycle. The tradeoff is characterized by a greater ion energy distribution magnitude and a lower ion energy, associated with increasing the duty cycle. As described, a voltage at the first electrode increases during the on period and decreases during the off period.

(58) The term “coupled” is used herein to refer to the direct or indirect coupling between two objects. For example, if object A physically touches object B and object B touches object C, then objects A and C may still be considered coupled to one another—even if objects A and C do not directly physically touch each other. For instance, a first object may be coupled to a second object even though the first object is never directly physically in contact with the second object.

(59) While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A method for plasma processing, comprising: determining a power associated with a first bias signal to be applied to a first electrode of a processing chamber, the power being determined based on a pressure inside a processing region of the processing chamber such that: the power s insufficient to generate a plasma inside the processing chamber by application of the power to the first electrode; and a second bias signal that exceeds the power when applied is sufficient to generate a monoenergetic peak; and performing a plasma-assisted etching process on processing a substrate using a plasma with a monoenergetic ion energy distribution, the plasma generated by: applying the first bias signal in accordance with the determined power to the first electrode; and applying the second bias signal to a second electrode of the processing chamber, wherein the second bias signal is configured to generate the plasma in the processing region and the first bias is applied while the second bias is applied.
2. The method of claim 1, wherein the power is determined based on the pressure for a particular gap inside the processing region of the processing chamber, the gap being between the first electrode and the second electrode or the first electrode and a grounded surface that defines at least a portion of the processing region.
3. The method of claim 1, further comprising applying a third bias signal to a third electrode of the processing chamber, wherein the first electrode and the third electrode are positioned within a substrate support assembly disposed within the processing region and a distance from a substrate supporting surface of the substrate support assembly.
4. The method of claim 1, wherein the first bias signal comprises a continuous wave (CW) signal.
5. The method of claim 1, wherein the power is further determined based on a gas composition disposed within the processing region of the processing chamber.
6. The method of claim 1, wherein the first electrode comprises an edge electrode of a substrate support assembly.
7. The method of claim 1, wherein the power is determined to be below a threshold such that the power is insufficient to generate the plasma inside the processing chamber.
8. The method of claim 1, wherein the second bias signal comprises a pulsed signal that comprises an on period and an off period, and a duty cycle associated with the pulsed signal is determined based on a tradeoff between ion energy and ion energy distribution magnitude associated with adjustments to the duty cycle.
9. The method of claim 8, wherein the tradeoff is characterized by a greater ion energy distribution

magnitude and a lower ion energy, associated with increasing the duty cycle.
10. The method of claim 1, wherein the second bias signal comprises a pulsed signal that comprises an on period and an off period, and wherein a power at the first electrode increases during the on period and decreases during the off period.
11. An apparatus for plasma processing, comprising: a memory having executable instructions; and a controller configured to execute the executable instructions and perform a plasma-enhanced deposition process on process a substrate using a plasma with a monoenergetic ion energy distribution, the plasma generated by: using a first source generator to apply a first bias signal to a first electrode of a processing chamber in accordance with a power determined based on a pressure inside a processing region of the processing chamber such that: the power is insufficient to generate a plasma inside the processing chamber by application of the power to the first electrode; and a second bias signal that exceeds the power when applied is sufficient to generate a monoenergetic peak; and using a second bias generator to apply the second bias signal to a second electrode of the processing chamber and generate the plasma.
12. The apparatus of claim 11, wherein the voltage and/or power is determined based on the pressure for a particular gap inside the processing region of the processing chamber, the gap being between the first electrode and the second electrode or the first electrode and a grounded surface that defines at least a portion of the processing region.
13. The apparatus of claim 11, wherein the controller is configured to cause a third bias generator to apply a third bias signal to a third electrode of the processing chamber, wherein the first electrode and the third electrode are positioned within a substrate support assembly disposed within the processing region and a distance from a substrate supporting surface of the substrate support assembly.
14. The apparatus of claim 11, wherein the first bias signal comprises a continuous wave (CW) signal.
15. The apparatus of claim 11, wherein the power is further determined based on a gas composition disposed within the processing region of the processing chamber.
16. The apparatus of claim 11, wherein the first electrode comprises an edge electrode of a substrate support assembly.
17. The apparatus of claim 11, wherein the power is determined to be below a power threshold such that the power is insufficient to generate the plasma inside the processing chamber.
18. The apparatus of claim 11, wherein the second bias signal comprises a pulsed signal that comprises an on period and an off period, and a duty cycle associated with the pulsed signal is determined based on a tradeoff between ion energy and ion energy distribution magnitude associated with adjustments to the duty cycle, wherein the tradeoff is characterized by a greater ion energy distribution magnitude and a lower ion energy, associated with increasing the duty cycle.